

Question - 1 (5 marks)

Write formal proof of the following theorem. Any sub-results from the theory of elementary matrices must be proven as Lemma before attempting the main proof (if needed in the main proof, i.e., mere assertions will be penalized).

Theorem: Let A and B be $n \times n$ matrices, then

$$\det(AB) = \det(A) \det(B)$$

Proof: If B is singular, then there exist a nonzero vector $\bar{x}$ such that

$$B\bar{x} = \bar{0}$$

$$\text{If } C = AB$$

$$\text{Then } C\bar{x} = AB\bar{x} = A\bar{0} = \bar{0}$$

This implies that C must also be singular.
and therefore

$$\det(AB) = 0 = \det(A) \det(B)$$

If B is nonsingular, B can be written as a product of elementary matrices.
i.e.

$$B = E_k E_{k-1} \dots E_2 E_1$$

$$\text{So } AB = A E_k E_{k-1} \dots E_2 E_1$$

$$\text{and } \det(AB) = \det(A E_k E_{k-1} \dots E_2 E_1)$$

Lemma :

Let E be an elementary matrix of type I
i.e. E can be created by interchanging any two rows of an identity matrix.

$$\text{Then } \det(EA) = \det(E) \det(A) \quad \text{--- (1)}$$

Proof of statement- (1) using mathematical induction.

For 2×2 matrices

$$\text{let } E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \& \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{23} \end{bmatrix}$$

$$EA = \begin{bmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{bmatrix}$$

$$\det(EA) = a_{21}a_{12} - a_{11}a_{22} = -(a_{11}a_{22} - a_{12}a_{21})$$

$$\det(EA) = -\det(A)$$

For $n > 2$: For the case $n = 3$

$$EA = \begin{bmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{bmatrix} \quad \begin{array}{l} \text{when 1st \& 3rd row} \\ \text{are interchanged.} \end{array}$$

$$\begin{aligned} \det(EA) &= -a_{21} \begin{vmatrix} a_{32} & a_{33} \\ a_{12} & a_{13} \end{vmatrix} + a_{22} \begin{vmatrix} a_{31} & a_{33} \\ a_{11} & a_{13} \end{vmatrix} - a_{23} \begin{vmatrix} a_{31} & a_{32} \\ a_{11} & a_{12} \end{vmatrix} \\ &= -a_{21}(a_{32}a_{13} - a_{33}a_{12}) + a_{22}(a_{31}a_{13} - a_{11}a_{33}) \\ &\quad - a_{23}(a_{31}a_{12} - a_{11}a_{32}) \\ &= a_{21}(a_{12}a_{33} - a_{13}a_{32}) - a_{22}(a_{11}a_{33} - a_{13}a_{31}) \\ &\quad + a_{23}(a_{11}a_{32} - a_{31}a_{22}) \\ &= -\det(A) \end{aligned}$$

In general if A is $n \times n$ & E_{ij} is $n \times n$ elementary formed by interchanging i th and j th row of I then

$$\text{In Particular } \det(E_{ij}; I) = -\det(I) = -1$$

$$\det(E_{ij}; A) = -\det(A)$$

Thus for any elementary matrix of type I

$$\det(EA) = \det(E) \det(A) = -\det(A)$$

let E be the elementary matrix of type II, which can be created by multiplying i th row of identity matrix by a non zero constant.

If determinant of EA is expanded by cofactors along i th row

Then,

$$\begin{aligned}\det(EA) &= \alpha a_{i1}A_{i1} + \alpha a_{i2}A_{i2} + \dots + \alpha a_{in}A_{in} \\ &= \alpha [a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}] \\ &= \alpha \det(A)\end{aligned}$$

In Particular

$$\det(EI) = \alpha \det(I) = \alpha$$

$$\det(EA) = \det(E) \det(A) = \alpha \det(A)$$

Let E be the elementary matrix of type III, which can be created by adding a constant multiple of i th row to the j th row of Identity matrix.

Since E is triangular and its diagonal elements are all 1, it follows

that- $\det(E) = 1$

If $\det(EA)$ is expanded by cofactors along the j th row, it follows from Lemma 2.2.1 that

$$\begin{aligned}\det(EA) &= (a_{j1} + ca_{i1})A_{j1} + (a_{j2} + ca_{i2})A_{j2} + \dots + (a_{jn} + ca_{in})A_{jn} \\ &= (a_{j1}A_{j1} + \dots + a_{jn}A_{jn}) + c(a_{i1}A_{j1} + \dots + a_{in}A_{jn}) \\ &= \det(A)\end{aligned}$$

Now,

$$\begin{aligned}\det(AB) &= \det(AE_k E_{k-1} \dots E_2 E_1) \\ &= \det(A) \det(E_k) \det(E_{k-1}) \dots \det(E_2) \det(E_1) \\ &= \det(A) \det(E_k \cdot E_{k-1} \dots E_2 \cdot E_1)\end{aligned}$$

$$\det(AB) = \det(A) \det(B) \quad \because B = E_k E_{k-1} \dots E_2 E_1$$

proved

Question - 2 (5 marks)

Write formal proof of the following theorem.

Theorem: Let A be a square matrix of order $n \times n$, then:

(i) $\det(\alpha A) = \alpha^n \det(A)$.

(ii) Furthermore, if A is non-singular; then, $\det(A^{-1}) = \frac{1}{\det(A)}$.

i/ **Proof:** let I be an $n \times n$ identity matrix.

Note that $\det(\alpha I) = \alpha^n$ since all the diagonal entries of αI are α and $\det(\alpha I)$ is the product of diagonal entries.

Now consider

$$\alpha A = \alpha I A$$

$$\text{and } \det(\alpha A) = \det(\alpha I A)$$

$$= \det(\alpha I) \det(A)$$

$$= \alpha^n \det(A) \quad \text{proved}$$

ii/ **Proof:** If A is a nonsingular matrix this implies

$$A A^{-1} = I = A^{-1} A$$

$$\text{and } \det(A A^{-1}) = \det(I) = \det(A^{-1} A)$$

$$\det(A) \det(A^{-1}) = 1$$

$$\boxed{\det(A^{-1}) = \frac{1}{\det(A)}}$$

proved

Question - 3 (5 marks)

Let $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ be vectors in a vector space V . If α and β are scalars, then prove the following results, and do mention which axiom(s) allow the proof to proceed to the next step.

- (a) If $\mathbf{x} + \mathbf{y} = \mathbf{x} + \mathbf{z}$, then $\mathbf{y} = \mathbf{z}$.
- (b) $\beta \mathbf{0} = \mathbf{0}$, for each scalar β .
- (c) If $\alpha \mathbf{x} = \mathbf{0}$, then either $\alpha = 0$, or $\mathbf{x} = \mathbf{0}$.

Let $\bar{x}$, $\bar{y}$, and $\bar{z}$ be any vectors in V and α and β are scalars.

a. Proof:

$$\text{If } \bar{x} + \bar{y} = \bar{x} + \bar{z}$$

For each vector $\bar{v}$ in a vector space there exist $-\bar{v}$ such that $\bar{v} + (-\bar{v}) = \bar{0} = (-\bar{v}) + \bar{v}$

This implies $\exists -\bar{x}$ such that $\bar{x} + (-\bar{x}) = \bar{0}$

$$\text{Therefore } \bar{x} + (-\bar{x}) + \bar{y} = \bar{x} + (-\bar{x}) + \bar{z}$$

$$\Rightarrow \bar{y} = \bar{z} \quad \text{proved}$$

b. Proof: If $\bar{y} = \beta \bar{0}$ then

$$\begin{aligned} \bar{y} + \bar{y} &= \beta \bar{0} + \beta \bar{0} \\ &= \beta (\bar{0} + \bar{0}) \\ &= \beta (\bar{0}) \\ &= \bar{y} \end{aligned}$$

$$\because \alpha(\bar{x} + \bar{y}) = \alpha\bar{x} + \alpha\bar{y}$$

it then follows that

$$\begin{aligned} \bar{y} + \bar{y} &= \bar{y} \\ (\bar{y} + \bar{y}) + (-\bar{y}) &= \bar{y} + (-\bar{y}) \\ \bar{y} + (\bar{y} + (-\bar{y})) &= \bar{y} + (-\bar{y}) \\ \bar{y} + \bar{0} &= \bar{0} \\ \bar{y} &= \bar{0} \end{aligned}$$

$$\because (\bar{x} + \bar{y}) + \bar{z} = \bar{x} + (\bar{y} + \bar{z})$$

$$\because \bar{x} + (-\bar{x}) = \bar{0} = (-\bar{x}) + \bar{x}$$

Hence

$$\bar{0} = \beta \bar{0} \quad \text{proved}$$

c/ Proof: If $\alpha \bar{x} = 0$ and $\alpha \neq 0$ then it follows from part (b) that-

$$\bar{0} = \frac{1}{\alpha} \bar{0} = \frac{1}{\alpha} (\alpha \bar{x})$$

$$\therefore \bar{0} = \alpha \bar{x}$$

$$= \left(\frac{1}{\alpha} \cdot \alpha \right) \bar{x}$$

$$\therefore \alpha(\beta \bar{x}) = (\alpha\beta) \bar{x}$$

$$= 1 \cdot \bar{x} = \bar{x}$$

$$\therefore 1 \cdot \bar{x} = \bar{x} \cdot 1 = \bar{x}$$

when 1 is scalar.

$$\Rightarrow \bar{x} = 0 \quad \text{proved}$$

Question - 4 (5 marks)

Consider a linear system whose augmented matrix is of the form:

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array} \right]$$

(a) For what values of a and b will the system have infinitely many solutions?

(b) For what values of a and b will the system be inconsistent?

Solution:

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array} \right]$$

a/ Applying row operations in order to reduce A to row echelon form
 $R_2 + (-1)R_1$ & $R_3 + (-1)R_1$

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & a-3 & b-2 \end{array} \right]$$

$R_3 + (-2)R_2$

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & a-5 & b-4 \end{array} \right]$$

when $a=5$ and $b=4$ then the given linear system will have a free variable and hence will have infinitely many solutions. Ans

b/ when $a=5$ and $b \neq 4$ then the coefficient matrix will have a row of 0's and the corresponding right hand side is nonzero. Hence the given system will be inconsistent. Ans

Question - 5 (5 marks)

A matrix A is called an *involution matrix* if $A^2 = I$, i.e., the matrix is its own inverse. A matrix B is said to be *idempotent* if $B^2 = B$.

- Give the algebraic rule for constructing an involution matrix. Using that rule construct an involution matrix A of order 4.
- Let B be an idempotent matrix. Show that $I + B$ is non-singular and $(I + B)^{-1} = I - \frac{1}{2}B$.

a/ Solution: Let A be an involution matrix, and let $\bar{u}$ be a unit vector then an algebraic rule for constructing an involution matrix is

$$A = I - 2\bar{u}\bar{u}^T \quad \text{Ans}$$

Construction of a 4×4 involution matrix:

Let I be an identity matrix of order 4×4 , let $\bar{u}$ be an arbitrary unit vector belonging to $\mathbb{R}^4$.

$$\bar{u} = (u_1, u_2, u_3, u_4)$$

Now

$$\bar{u}\bar{u}^T = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \begin{bmatrix} u_1 & u_2 & u_3 & u_4 \end{bmatrix}$$

$$\bar{u}\bar{u}^T = \begin{bmatrix} u_1^2 & u_1u_2 & u_1u_3 & u_1u_4 \\ u_1u_2 & u_2^2 & u_2u_3 & u_2u_4 \\ u_1u_3 & u_2u_3 & u_3^2 & u_3u_4 \\ u_1u_4 & u_2u_4 & u_3u_4 & u_4^2 \end{bmatrix}$$

Now Since

$$A = I - 2\bar{u}\bar{u}^T$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} u_1^2 & u_1 u_2 & u_1 u_3 & u_1 u_4 \\ u_1 u_2 & u_2^2 & u_2 u_3 & u_2 u_4 \\ u_1 u_3 & u_2 u_3 & u_3^2 & u_3 u_4 \\ u_1 u_4 & u_2 u_4 & u_3 u_4 & u_4^2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1-2u_1^2 & 2u_1 u_2 & 2u_1 u_3 & 2u_1 u_4 \\ 2u_1 u_2 & 1-2u_2^2 & 2u_2 u_3 & 2u_2 u_4 \\ 2u_1 u_3 & 2u_2 u_3 & 1-2u_3^2 & 2u_3 u_4 \\ 2u_1 u_4 & 2u_2 u_4 & 2u_3 u_4 & 1-2u_4^2 \end{bmatrix}$$

Ans

✓ Proof: Let B be an idempotent matrix i.e.
 $B^2 = B$

Then

$$\begin{aligned} (I+B)\left(I - \frac{1}{2}B\right) &= I^2 - \frac{1}{2}IB + BI - \frac{1}{2}B^2 \\ &= I - \cancel{\frac{1}{2}B} + \cancel{B} - \cancel{\frac{1}{2}B} = I \quad \because B^2 = B \end{aligned}$$

$$\begin{aligned} \text{and } \left(I - \frac{1}{2}B\right)(I+B) &= I^2 + IB - \frac{1}{2}BI - \frac{1}{2}B^2 \\ &= I + B - \frac{1}{2}B - \frac{1}{2}B = I \end{aligned}$$

$$\therefore (I+B)\left(I - \frac{1}{2}B\right) = \left(I - \frac{1}{2}B\right)(I+B) = I$$

Therefore $(I+B)$ is nonsingular and $(I+B)^{-1} = I - \frac{1}{2}B$

proved

Question - 6 (5 marks)

Let A and B be $n \times n$ matrices and define $2n \times 2n$ matrices S and M by

$$S = \begin{bmatrix} I & A \\ O & I \end{bmatrix}, \quad M = \begin{bmatrix} AB & O \\ B & O \end{bmatrix}$$

(a) Compute the block form of S^{-1} .

(b) Use the block form of S^{-1} (from 6a) to compute the block form of $S^{-1}MS$.

Solution: Let A and B be $n \times n$ matrices and define $2n \times 2n$ matrix S by

$$S = \begin{bmatrix} I & A \\ O & I \end{bmatrix}$$

To find S^{-1} augment S with I of order $2n$.

$$[S|I] = \left[\begin{array}{cccc|cccc} 1 & 0 & \dots & 0 & a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ 0 & 1 & & \vdots & a_{21} & a_{22} & & \vdots & 0 & 1 & & \vdots \\ \vdots & & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & a_{n1} & a_{n2} & \dots & a_{nn} & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 1 & 0 & \dots & 0 & 0 & \vdots & \vdots & \vdots & \vdots \\ \vdots & & \vdots & 0 & 1 & & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & 0 & 0 & 0 & \dots & 0 & 1 & 0 & 0 & \dots & 1 \end{array} \right]$$

Apply row operations to reduce S into reduced row echelon form (i.e. into identity), identity part of the augmented system will become S^{-1} .

Clearly, we can see that-

$$[S|I] = \left[\begin{array}{cc|cc} I & O & I & -A \\ O & I & O & I \end{array} \right]$$

$$\Rightarrow S^{-1} = \begin{bmatrix} I & -A \\ O & I \end{bmatrix}$$

Ans

$$\hookrightarrow S^{-1}MS = \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I \end{bmatrix}$$

$$= \begin{bmatrix} I & -A \\ 0 & I \end{bmatrix} \begin{bmatrix} AB & ABA \\ B & BA \end{bmatrix}$$

$$S^{-1}MS = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix}$$

Ans